Atherosclerosis 394 (2024) 117170

Contents lists available at ScienceDirect

Atherosclerosis

journal homepage: www.elsevier.com/locate/atherosclerosis

Obstructive sleep apnea, coronary calcification and arterial stiffness in patients with diabetic kidney disease

Sebastian Nielsen [a,*], Jakob Nyvad [a], Kent Lodberg Christensen [b], Per Løgstrup Poulsen [c,d], Esben Laugesen [c,e], Erik Lerkevang Grove [b,d], Niels Henrik Buus [a,d]

[a] Department of Renal Medicine, Aarhus University Hospital, Aarhus, Denmark
[b] Department of Cardiology, Aarhus University Hospital, Aarhus, Denmark
[c] Steno Diabetes Center, Aarhus University Hospital, Denmark
[d] Department of Clinical Medicine, Faculty of Heath, Aarhus University, Aarhus, Denmark
[e] Diagnostic Center, Silkeborg Regional Hospital, Silkeborg, Denmark

ARTICLE INFO

Keywords:
Obstructive sleep apnea
Diabetic kidney disease
Coronary calcification
Arterial stiffness
Coronary agatston score

ABSTRACT

Background and aims: Obstructive sleep apnea (OSA) may accelerate arterial calcification, but the relation remains unexplored in diabetic kidney disease (DKD). We examined the associations between OSA, coronary calcification and large artery stiffness in patients with DKD and reduced renal function.

Methods: Patients with type 2 diabetes, estimated glomerular filtration rate (eGFR) < 60 ml/min/1.73 m^2 and urine albumin-creatinine ratio (UACR) > 30 mg/g were tested for OSA quantified by the apnea-hypopnea index (AHI, events/hour). Patients without OSA (AHI$<$ 5) were compared to patients with moderate (AHI 15–29) or severe (AHI $\geq$30) OSA and underwent computed tomography angiography with coronary Agatston scoring (CAS) to quantify coronary calcification. Arterial stiffness was determined as carotid-femoral pulse wave velocity (PWV).

Results: Among 114 patients with acceptable AHI recordings had 43 no OSA, 33 mild OSA and 38 moderate-severe OSA. Mean age of the 74 patients completing the study was 71.5 $\pm$ 9.4 years (73% males), eGFR 32.2 $\pm$ 12.3 ml/min/1.73 m^2 and UACR 533 (192–1707) mg/g. CAS (ln-transformed) was significantly higher in patients with OSA compared to patients without (6.6 $\pm$ 1.7 *vs.* 5.6 $\pm$ 2.4, $p = 0.04$), and the same was observed for PWV (11.9 $\pm$ 2.7 *vs.* 10.5 $\pm$ 2.2 m/s, $p = 0.02$). In multivariable linear regression analyses adjusted for sex, age, body mass index, UACR, and mean arterial pressure, moderate-severe OSA remained significantly associated with PWV but not with CAS. Dominance analysis revealed OSA as the third and second most important factor relative to CAS and PWV respectively.

Conclusions: In DKD patients, moderate-severe OSA is a significant predictor of arterial stiffness but is not independently associated with coronary calcification.

1. Introduction

Diabetic kidney disease (DKD) due to type 2 diabetes is the leading cause of chronic kidney disease (CKD) [1]. DKD is associated with increased morbidity and mortality, mainly due to cardiovascular disease with accelerated vascular calcification as an important contributor [2,3]. However, despite strict treatment of classical risk factors like hypertension, dyslipidaemia, and hyperglycaemia, calcification tends to progress at an increased rate in DKD patients, rendering this population at an excessively high cardiovascular risk [3,4].

Previous observations have linked development of cardiovascular disease to the presence of obstructive sleep apnea (OSA) [5,6]. OSA is characterized by frequent episodes of complete (apnea) or partial (hypopnea) upper airway obstruction during sleep leading to hypoxemia, arousal, sleep fragmentation and increased sympathetic activity [7]. These obstructive events are associated with systemic inflammation, oxidative stress, endothelial dysfunction facilitating coronary atherosclerosis and arteriosclerosis linking OSA to a more severe cardiovascular risk profile with increased risk of events [7–10]. Identification of OSA in high-risk patients may potentially add further risk

* Corresponding author. Department of Renal Medicine, Aarhus University Hospital, Palle Juul-Jensens Boulevard 99, 8200, Aarhus N, Denmark.
E-mail address: Sebane@rm.dk (S. Nielsen).

https://doi.org/10.1016/j.atherosclerosis.2023.06.076
Received 6 March 2023; Received in revised form 14 June 2023; Accepted 15 June 2023
Available online 29 June 2023

information that may guide intensified prophylactic cardiovascular treatment [5]. Nevertheless, OSA remains undiagnosed and under-treated in clinical practice [11].

OSA is common among type 2 diabetes patients with a reported prevalence ranging from 23% to 87% [12]. This is at least the double of the prevalence in the general population in persons of comparable age and gender where it has been estimated to be between 9% and 38% [12]. OSA is also more prevalent in patients with chronic kidney disease (CKD), and the risk increases with the severity of CKD. Specifically, the reported prevalence of OSA in patients with normal kidney function is around 27%, while it increases to 41% in CKD patients and 57% in those with end-stage kidney disease [13]. The prevalence of OSA is very high in patients with DKD reaching approximately 50% [14].

The combination of diabetes and reduced kidney function could therefore lead to a very high risk of OSA. However, the association to OSA has only been assessed in one previous study in patients with reduced glomerular filtration rate and proteinuria and the relation between OSA and coronary calcification in DKD patients has to our knowledge never been examined [15]. The aim of the present study was therefore to investigate the association between OSA, coronary calcification and large arterial stiffness in high-risk patients with DKD.

2. Patients and methods

2.1. Design and study population

We conducted an observational cross-sectional study in patients with DKD to examine the association between OSA and coronary artery calcification (CAC) and large artery vascular stiffness. The primary study outcome was CAC quantified by coronary Agatston score (CAS) and arterial stiffness based on carotid-femoral pulse wave velocity (PWV) as the secondary outcome. Participants were enrolled between September 2020 and January 2022 from the outpatient clinics for renal medicine or diabetes at Aarhus University Hospital or from primary health care within the local region.

Inclusion criteria were age above 18 years, type 2 diabetes based on present antidiabetic treatment and an ICD diagnosis code of diabetes, urine albumin-creatinine ratio (UACR) above 30 mg/g in at least 2 urine samples within the last 12 months, estimated glomerular filtration rate (eGFR) 10–59 ml/min/1.73 m^2 (but not receiving renal replacement treatment) in at least 2 out of 3 separate measurements over a period of 12 months and the ability to give informed consent. Exclusion criteria were continuous positive airway pressure (CPAP) treatment within the last 3 months before recruitment, previous coronary artery bypass grafting (CABG) and/or previous percutaneous coronary intervention (PCI).

Eligible patients were screened for OSA (see below) and excluded if meeting the following criteria: less than 4 h of recording, mild OSA, more than 50% central apnea events within the OSA group or the presence of Cheyne-Stokes respiration. Patients not meeting these exclusion criteria continued in the study with coronary computed tomography angiography (CTA) and measurement of PWV performed within two months of the OSA assessment. Office brachial blood pressure (BP) was determined three times with a fully automated oscillometric device (Microlife A2/B2 Basic, Microlife AG Swiss Corporation, Widnau, Switzerland) after being seated for at least 10 min. Venous blood and urine samples were collected on the day of the OSA evaluation.

The study was approved by the Danish Data Protection Agency (1-16-02-397-20), the Central Denmark Region Committees on Health Research Ethics (1-10-72-122-20) and registered at clinicaltrials.gov (NCT04549324). The study was conducted according to the declaration of Helsinki. All patients gave written informed consent before inclusion in the study.

2.2. Determination of sleep apnea

A home sleep test for OSA was performed using the validated ApneaLink+ (ResMed Corp., CA, USA) [16,17], which is a portable three-channel device measuring nasal airflow, respiratory effort, and peripheral oxygen saturation. Participants were provided with oral and written instructions for mounting the device before sleep with the opportunity to contact an investigator if any technical issue occurred. If recording time was less than 4 h, the patient was offered to sleep with ApneaLink+ for one more night, and if the recording period remained under 4 h, the patient was excluded. The recordings were analysed by the ApneaLink+ software and OSA severity was quantified by the apnea-hypopnea index (AHI) defined as the average number of apnea and hypopnea events per hour of sleep [18]. An apnea was defined according to standardized settings as interruption in airflow for at least 10 s and a hypopnoea as a decrease in airflow for at least 10 s combined with ≥3% desaturation compared to pre-event baseline [19]. Absence of OSA was defined as an AHI below 5 events/hour, mild OSA as 5–14 events/hour, moderate OSA as 15–29 events/hour and severe OSA as ≥ 30 events/hour.

2.3. Coronary CT angiography acquisition and analysis

Coronary CTA was performed using a dual-source CT scanner (SOMATOM Definition Force; Siemens, Forcheim, Germany) according to practice guidelines from the Society of Cardiovascular Computed Tomography [20]. If necessary, oral and/or intravenous beta-blockers were administered, targeting a heart rate below 60 beats per minute. Sublingual nitro-glycerine spray 0.8 mg was administered 3–5 min before the scan. An initial non-contrast scan for CAS was performed using high-pitch spiral acquisition mode and 120 kV scan protocols with prospective electrocardiographic triggering were applied [21,22]. Data acquisition was performed with 100 kV tube voltage in patients weighing ≤70 kg, and 120 kV in patients weighing >70 kg. Filtered back-projection was used for image reconstruction in all patients and the scans were assessed using axial images and multiplanar reconstructions.

The presence of CAC was defined as a CAS > 0 [23]. CAS severity was grouped into predefined categories of 0, 1–100, 101–300 and ≥301 based on Greenland et al. who combined CAS and the Framingham risk score for cardiovascular risk prediction [23,24].

The percentile of CAS was estimated using the scoring software of the Multi-Ethnic Study of Atherosclerosis (MESA), which included the variables of age, sex, ethnicity and CAS [25]. All coronary CTAs were evaluated and scored by the same experienced cardiologist (ELG) who was blinded to OSA status.

2.4. Pulse wave velocity

Arterial stiffness was determined by PWV using SphygmoCor© (version 9, Atcor Medical, Sydney, Australia) in accordance to recent recommendations [26]. Brachial BP was measured three times with the fully automated oscillometric device (Microlife WatchBP Office, Microlife AG, Widnau, Switzerland) after 5 min of supine rest. The travel distance was measured as a straight line from the right carotid artery to the right femoral artery multiplied by 0.8 [26]. The mean PWV of two measurements was used for each participant. We defined PWV as increased if the velocity was ≥9.6 m/s based on the clinical cut-off value of elevated cardiovascular risk [26].

2.5. Statistical analysis

All variables were tested for normal distribution by visual inspection of histograms and QQ-plots. Continuous variables with normal distribution are presented as mean ± standard deviation or as median with interquartile range (IQR) for data not normally distributed. Data are stratified by OSA-severity (AHI < 5/hour and ≥15/hour). Means of continuous variables were compared with an independent-sample *t*-test and skewed variables with Wilcoxon Rank test to compare medians. Chi-squared test was used to compare categorical variables and presented as number and percentage.

CAS data were logarithmic transformed due to asymmetric distribution. All raw CAS data was added up by 1 (lnCAS+1]). Mean arterial pressure (MAP) was calculated as diastolic BP + 1/3 of the pulse pressure.

Statistical analyses were carried out using both linear regression analysis for CAS and PWV and logistic regression analysis for the categorical variable of PWV $<$ or $\geq$9.6 m/s and CAS $\leq$300 or $>$ 300. Multivariable linear and logistic regressions were performed to adjust for cardiovascular risk factors such as sex, age, body mass index (BMI), UACR and MAP. Dominance analysis was used to estimate the relative importance of the cardiovascular risk factors which were also included the multivariable regression models [27]. The residuals of the regression models were examined with QQ-plots. Two-sided *p*-values $<$ 0.05 were defined as statistically significant. All analyses were run by STATA 17.0 software for Windows (StataCorp LP, College State, TX).

3. Results

3.1. Clinical characteristics

A total of 510 patients were invited by letter to participate in the study of whom 106 refused and 212 did not respond (Fig. 1). The majority of the non-responders were from the primary health care and mainly men. A total of 120 patients (24%) were examined by ApneaLink+ with 114 having sufficient data. Of these had 43 patients (38%) no OSA, 33 (29%) mild OSA and 38 (33%) moderate-severe OSA. Thirteen (11%) patients discontinued the study after enrolment due to inadequate ApneaLink+data (n = 4), withdrawal of consent (n = 7) or coronary intervention before CTA (n = 2), leaving 74 patients for the final analysis.

Table 1 shows clinical characteristics of the study population. Patients with moderate-severe OSA had significantly higher body mass index (BMI) and were more likely males as compared to patients without OSA. Age, smoking habits, eGFR, diabetes duration and BP were comparable in patients with and without OSA. HbA1c, total cholesterol and triglycerides were slightly higher in patients without OSA. The median level of UACR was higher in the OSA group, but based on Kidney Disease Improving Global Outcomes (KDIGO) CKD staging, the numbers of patients with and without OSA within each category were quite similar (Supplementary Table S1). The use of glucose- and lipid-lowering

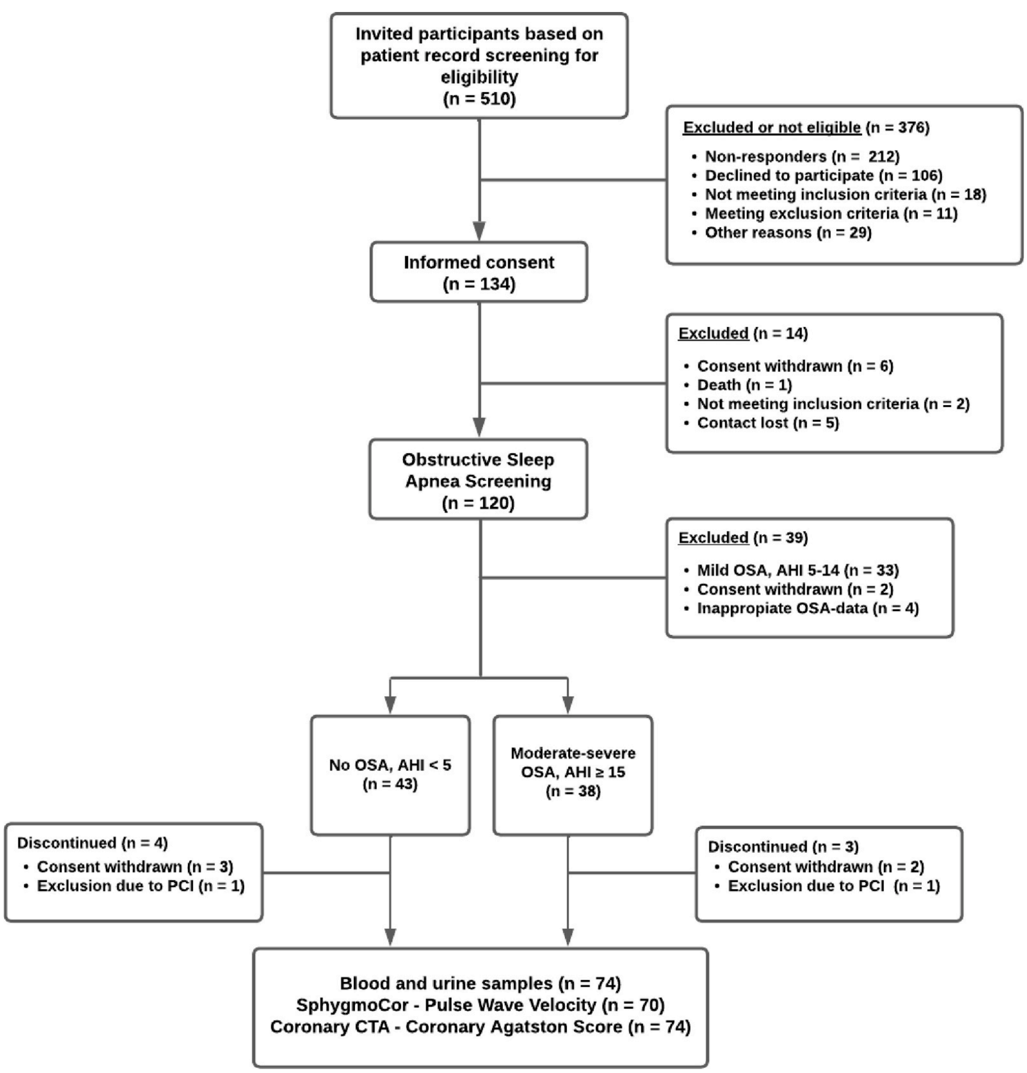

Fig. 1. Flow chart of the study population.
OSA: Obstructive sleep apnea; AHI: Apnea-hypopnea index; PCI: Percutaneous coronary intervention; Coronary CTA: Coronary computed tomography angiography.

Table 1
Baseline characteristics of the study population according to obstructive sleep apnea status.

	Total population (n = 74)	No OSA (n = 39)	OSA (n = 35)	*p*-value*
Age (years)	71.5 ± 9.4	71.4 ± 10.3	71.7 ± 8.5	0.89
Males, n (%)	54 (73.0)	24 (61.5)	30 (85.7)	0.02
Body mass index (kg/m^2)	31.3 ± 5.6	29.5 ± 5.0	33.3 ± 5.5	<0.01
Smoking status, n (current/former/never)	9/33/32	5/16/18	4/17/14	0.80
Duration of diabetes (years)	18.1 ± 8.4	17.3 ± 8.8	18.9 ± 8.1	0.42
Office systolic blood pressure (mmHg)	140.6 ± 16.5	140.9 ± 17.5	140.3 ± 15.6	0.89
Office diastolic blood pressure (mmHg)	78.5 ± 10.0	79.6 ± 10.4	77.3 ± 9.4	0.31
Apnea-hypopnea index	4 (2–25)	2 (1–4)	29 (20–41)	<0.0001
Laboratory values				
eGFR (ml/min/1.73 m^2)	32.2 ± 12.3	32.5 ± 11.4	31.9 ± 13.3	0.84
Urine albumin-creatinine ratio (mg/g)	533 (192–1707)	409 (131–897)	941 (263–2414)	0.046
Hemoglobin A1c (mmol/mol)	59.6 ± 14.4	62.8 ± 14.3	56.0 ± 13.8	0.042
Total cholesterol (mmol/l)	3.98 ± 0.99	4.23 ± 0.99	3.71 ± 0.93	0.024
HDL cholesterol (mmol/l)	1 (0.89–1.2)	1 (0.89–1.2)	1 (0.89–1.2)	0.97
Triglycerides (mmol/l)	1.9 (1.2–2.65)	2.2 (1.3–3.4)	1.6 (1.1–2.3)	0.028
LDL cholesterol (mmol/l)	1.94 ± 0.86	2.05 ± 0.70	1.82 ± 1.00	0.27
Medication				
Antihypertensive treatment, n (%)	68 (91.9)	36 (92.3)	32 (91.4)	0.89
Insulin, n (%)	47 (63.5)	25 (64.1)	22 (62.9)	0.91
Metformin, n (%)	29 (39.2)	13 (33.3)	16 (45.7)	0.28
GLP-1 analogue, n (%)	26 (35.1)	14 (35.9)	12 (34.3)	0.88
DDP-4 inhibitor, n (%)	15 (20.3)	6 (15.4)	9 (25.7)	0.27
SGLT-2 inhibitor, n (%)	17 (23.0)	10 (25.6)	7 (20.0)	0.56
Sulphonylurea, n (%)	4 (5.4)	3 (7.7)	1 (2.9)	0.36
Lipid-lowering treatment, n (%)	67 (90.5)	35 (89.7)	32 (91.4)	0.81
Comorbidity				
Atrial fibrillation, n (%)	21 (28.4)	9 (23.1)	12 (35.6)	0.29
Chronic obstructive lung disease, n (%)	11 (14.9)	5 (12.8)	6 (17.1)	0.60

Data are presented as mean ± SD or medians with interquartile range. *Moderate-severe OSA compared to no-OSA. OSA: obstructive sleep apnea; eGFR: estimated glomerular filtration rate; GLP-1: glucagon like peptide 1; DDP-4: dipeptidyl peptidase-4; SGLT-2: sodium-glucose co-transporter type-2.
Missing data by category: Triglycerides 2, 0, 2; LDL cholesterol 7, 2, 5.

medication did not differ between the two groups.

3.2. Coronary calcification

Coronary calcification was present in 97% of patients with moderate-severe OSA and 92% of patients without OSA with the crude CAS data being presented in Fig. 2A and categorized CAS severity in Fig. 2B. CAS (ln-transformed) was significantly higher in patients with OSA as compared to patients without OSA (6.6 ± 1.7 *vs.* 5.6 ± 2.4, $p = 0.04$). Furthermore, the mean percentile of CAS was 79 in patients with OSA as compared to 69 in patients without OSA.

In an unadjusted model, CAS was significantly associated with the presence of OSA and male sex while the association to age was borderline significant (model 1 in Supplementary Table S2). In a multivariable linear regression model with adjustment for sex, age, BMI, UACR and MAP, the association between CAS and OSA became insignificant (β = 0.440, 95% CI [−0.65, 1.53]), whereas CAS remained significantly associated to male sex and age (model 2 in Supplementary Table S2). In the dominance analysis, male sex was the most relative importance to CAS with age and OSA being second and third, respectively (Fig. 4A).

In the univariate logistic regression model, a CAS > 300 was associated with increased risk of OSA (OR 4.14, 95% CI [1.41, 12.21]). However, in the multivariable analysis the association was not statistically significant (OR 2.79, 95% CI [0.76–10.28]) (Supplementary Table S2B).

3.3. Arterial stiffness

PWV measurements could be obtained in 70 (95%) of the enrolled patients. Missing data (n = 4) were mainly the result of obesity. Mean PWV was significantly higher in patients with OSA compared to patients without OSA (11.9 ± 2.7 *vs.* 10.5 ± 2.2, $p = 0.02$, Fig. 3A) and significantly more patients with OSA had PWV ≥9.6 m/s (84% *vs.* 61%, $p = 0.0353$, Fig. 3B) as compared to patients without OSA. Neither systolic BP (138.7 ± 17.3 *vs.* 138.9 ± 16.4 mmHg, $p = 0.96$) nor diastolic BP (75.2 ± 10.3 *vs.* 74.8 ± 8.5, $p = 0.86$) at the PWV examination differed between patients with and without OSA.

In an univariable linear regression analysis, the presence of OSA as well as MAP was associated with PWV (model 1 in Supplementary Table S3A). In multivariable linear regression both OSA (β = 1.442, 95% CI [0.16, 2.72]) and MAP (β = 0.089, 95% CI [0.03, 0.15]) remained significantly associated with PWV (model 2 in Supplementary Table S3A). MAP was estimated with the highest importance relative to PWV with OSA being the second most important factor (Fig. 4B).

Also, in a logistic regression analysis, a PWV ≥9.6 m/s was associated with increased risk of OSA (OR 4.43, 95% CI [1.05, 18.67]) even after multiple adjustments (Supplementary Table S3B).

The primary findings of the study are illustrated on Fig. 5.

4. Discussion

To the best of our knowledge, the present study is the first to examine the association between OSA, CAC and arterial stiffness in patients with DKD. In our population approximately two thirds had OSA and patients with moderate-severe OSA had more pronounced CAC. Moderate-severe OSA was significantly associated with both CAS and PWV, although only the association to PWV remained significant after adjustments for cardiovascular risk factors.

4.1. OSA, coronary artery disease and coronary calcification

OSA is frequent in patients diagnosed with coronary heart disease with a reported prevalence of moderate or severe OSA of 38–65% [28]. Shah et al. found that patients above 50 years with mild OSA had a 2-fold increased risk for cardiovascular events or death related to coronary calcification and plaque formation [29]. In addition, patients with severe OSA were found to have a lower 18-month event-free survival after a ST-segment elevation myocardial infarction [30]. These observations suggest OSA as an important factor for cardiovascular disease.

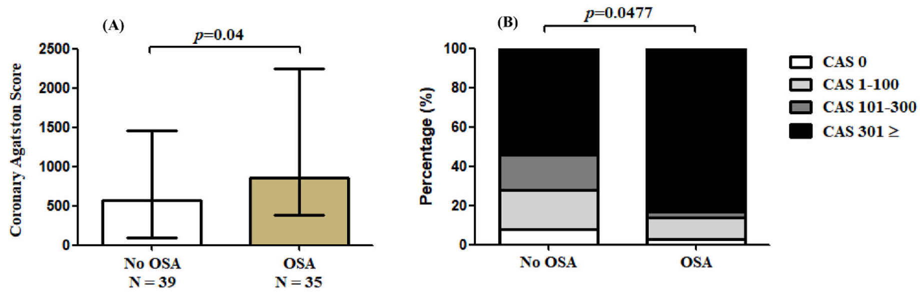

Fig. 2. Obstructive sleep apnea and coronary Agatston score.
(A) Crude CAS stratified on OSA-status (median with interquartile range). The *p*-value is calculated using the independent *t*-test on ln-transformed CAS data. (B) Distribution of categorized CAS stratified on OSA status *p*-value based on chi-squared test). OSA: Obstructive sleep apnea; CAS: Coronary Agatston score.

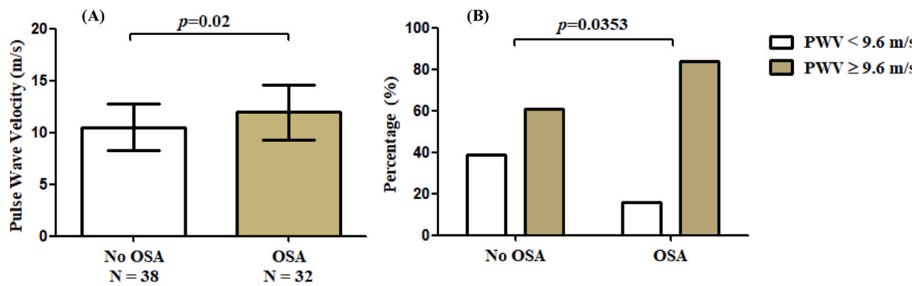

Fig. 3. Obstructive sleep apnea and carotid-femoral pulse wave velocity.
(A) PWV stratified according to OSA status (mean ± SD). (B) Percentage of participants with PWV < 9.6 and ≥9.6 m/s respectively according to OSA status (*p*-value based on chi-squared test). OSA: Obstructive sleep apnea; PWV: carotid-femoral pulse wave velocity.

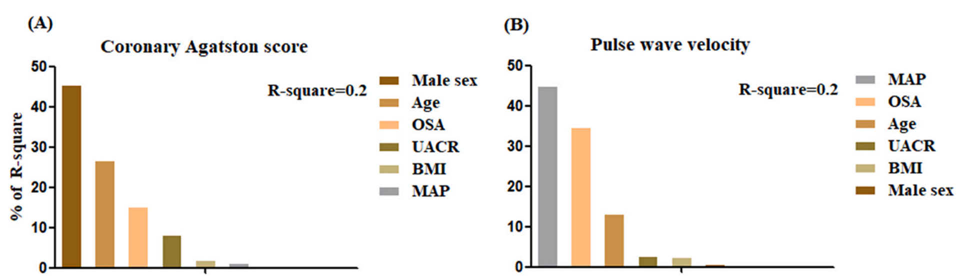

Fig. 4. Relative importance of cardiovascular risk factors for (A) coronary Agatston score and (B) pulse wave velocity.
OSA: Obstructive sleep apnea; BMI: body mass index; UACR: urine albumin-creatine ratio; MAP: mean arterial pressure.

The potential importance of diagnosing OSA to improve cardiovascular risk prediction and the possibility of treating the condition to reduce risk was recently addressed by a statement from the American Heart Association [10].

The pathophysiological links between OSA and vascular disease are not clearly identified. Several observational studies have evaluated the association between OSA and CAC rendering different results. Some report a higher rate of CAC and a significant association to CAS in patients with OSA [31–33]. Weinreich et al. reported a doubling of AHI to be associated with a 19% increase in CAS in men ≤65 years [31], and also Sorajja et al. found OSA to be independently associated with CAC after adjusting for traditional coronary artery disease risk factors [33]. In other studies, the association between OSA and CAS disappeared after adjustment for BMI [34–36].

The independent contribution of OSA to CAC severity has not previously been evaluated in a high-risk group such as DKD. In our study, only including DKD patients with reduced eGFR, CAC was very frequent, and the CAS was even higher than previously reported in DKD patients with eGFR of a similar range [37,38]. We believe the difference between the findings in these studies and ours can be partly explained by the higher mean age of our cohort and possibly also a higher prevalence of OSA. Based on the calculated CAS percentile and the Framingham risk score, patients without OSA will have a 10-year risk of 10–20% and patients with moderate-severe OSA a 10-year risk of cardiovascular disease of more than 20% [39]. However, using multivariable regression the influence of OSA on CAC weakened and became insignificant with the dominance analysis still pointing to male sex as the most important contributor to CAC.

However, OSA *per se* contributes more than other cardiovascular risk factors as UACR and MAP in the dominance analysis. Even though BMI, as expected, was higher in the OSA group, our statistical analysis did not point to any considerable contribution of this factor to CAS. Interestingly, we observed that patients without OSA and patients with moderate-severe OSA were comparable regarding most cardiovascular risk factors including age, smoking habits, eGFR, diabetes duration and BP. LDL cholesterol and HbA1c even seemed better controlled in the OSA patients. Of interest the albuminuria level was more than double in the OSA group and the relative weight of this factor was estimated to 8.3% relative to CAS emphasizing the importance of albuminuria as a vascular risk factor in CKD.

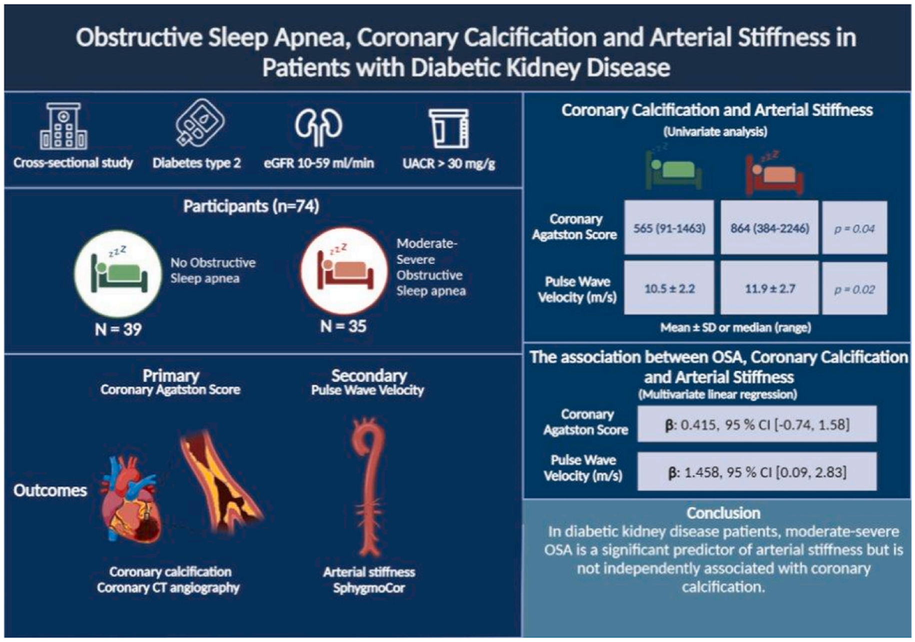

Fig. 5. Graphical abstract illustrating the study design and primary findings regarding the relationships among obstructive sleep apnea, coronary calcification, and arterial stiffness in patients with diabetic kidney disease.
eGFR: estimated glomerular filtration rate; UACR: urine albumin-creatine ratio; OSA: obstructive sleep apnea.

4.2. OSA and large artery stiffness

We found OSA to be significantly associated with PWV independent of other factors including age and BP which are usually considered most important for vascular stiffness. This observation is in line with previous studies [40–42], and demonstrates how OSA seems to damage the vascular wall. Interestingly, our analysis showed that the relative importance of OSA associated to PWV was larger than age.

There are multiple potential mechanisms underlying this including systemic inflammation, oxidative stress, increased sympathetic activity and endothelial dysfunction which facilitate arterial calcification [43]. A previous Japanese study demonstrated severe OSA to be associated with a higher extent of calcification in the abdominal aorta based on Agatston scoring, but the association weakened after adjustment for other risk factors [44]. A more recent study showed an independent association between OSA and Agatston score of the thoracic aorta [45]. We did not quantify aortic calcification, but the higher PWV in patients with OSA are in line with these observations.

In accordance with recent recommendations, PWV >9.6 m/s can be used as cut-off for an increased risk of cardiovascular events [26,46]. Our data indicate that DKD patients with OSA more often have a PWV above this level despite seemingly well-controlled BP and other known risk factors. The PWV data may therefore support the CAS data concerning risk stratification suggesting that DKD patients with OSA are at a higher risk level.

4.3. Clinical implications

With OSA being very frequent in DKD patients, treatment options should be carefully considered. Weight loss will reduce the severity of OSA in some patients, but CPAP is more effective to alleviate symptoms. The effect of CPAP on vascular damage, such as coronary or aortic calcification, is largely unknown and CPAP perhaps only to a limited extent affect the calcification processes [10]. A recent Danish study found no effect of CPAP treatment on arterial stiffness in obese diabetic patients [47]. This may suggest that traditional risk factors as well as CKD-bone and mineral disorders should be treated earlier and more intensively in DKD with OSA [48]. Interestingly, Wojeck et al. showed in a recent study that the use of sodium-glucose cotransporter 2 inhibitor (SGLT2i) treatment halved the incidence of OSA in patients with type 2 diabetes and atherosclerotic CVD [49]. This study may suggest that treating DKD patients with SGLT2i could have a beneficial effect on the treatment and/or prevention of OSA.

4.4. Limitations

We recruited most participants from a renal or diabetes clinic and not from clinics specialized in sleep disorders. Accordingly, our study cohort can be expected to resample other hospital based DKD populations stressing OSA as a common problem in this patient category. We excluded patients with prior CABG and PCI to improve the precision of coronary artery Agatston scoring and most of our patients had no previous coronary events. The extent of OSA in previously revascularized DKD patients is unknown. Due to the cross-sectional study design, we cannot infer a direct causality between OSA and CAS or large artery stiffness, but the adjusted analyses may suggest an important contribution. Additionally, it is important to note that residual confounding from unknown variables may still be present despite the use of multivariable analyses.

We used a portable sleep device and not polysomnography, which is considered a gold standard in the diagnosis of OSA. However, ApneaLink + has been validated against polysomnography in both obese and non-obese patients and ApneaLink + simplifies OSA diagnosis in clinics not specialized in sleep disorders [17,50].

The general frailty of elderly DKD patients led to many non-responders or patients declining participation and the prevalence of OSA in these is unknown. However, age and sex distribution of those not participating was similar to the patients willing to participate.

4.5. Conclusion

Patients with DKD and moderate-severe OSA have more pronounced CAC and higher carotid-femoral PWV than patients without OSA. Moderate-severe OSA is the third and second most important factor relative to CAS and PWV, respectively. However, in the multivariable analyses, the association with CAS was weakened, and only carotid-femoral PWV remained independently associated with OSA. The higher burden of vascular calcification in OSA may contribute to an increased cardiovascular risk and calls for studies exploring whether more extensive intervention/medication of risk factors is warranted.

Financial support

The study was supported by the Independent Research Fund Denmark, Karl G Andersen's Foundation and the Danish Society of Nephrology.

CRediT authorship contribution statement

Sebastian Nielsen: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Visualization, Project administration, Funding acquisition. **Jakob Nyvad:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Kent Lodberg Christensen:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Per Løgstrup Poulsen:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Esben Laugesen:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Erik Lerkevang Grove:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Niels Henrik Buus:** Conceptualization, Methodology, Formal analysis, Writing – original draft, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank all clinical staff at the outpatient clinics of Department of Renal medicine and Steno Diabetes Centre Aarhus at Aarhus University Hospital, for their support.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.atherosclerosis.2023.06.076.

References

[1] K.R. Tuttle, et al., Diabetic kidney disease: a report from an ADA Consensus Conference, Diabetes Care 37 (10) (2014) 2864–2883.
[2] J. Chen, et al., Coronary artery calcification and risk of cardiovascular disease and death among patients with chronic kidney disease, JAMA Cardiol 2 (6) (2017) 635–643.
[3] C.S. Fox, et al., Associations of kidney disease measures with mortality and end-stage renal disease in individuals with and without diabetes: a meta-analysis, Lancet 380 (9854) (2012) 1662–1673.
[4] M. Tonelli, et al., Risk of coronary events in people with chronic kidney disease compared with those with diabetes: a population-level cohort study, Lancet 380 (9844) (2012) 807–814.
[5] J.M. Marin, et al., Long-term cardiovascular outcomes in men with obstructive sleep apnoea-hypopnoea with or without treatment with continuous positive airway pressure: an observational study, Lancet 365 (9464) (2005) 1046–1053.
[6] C.D. Turnbull, Intermittent hypoxia, cardiovascular disease and obstructive sleep apnoea, J. Thorac. Dis. 10 (Suppl 1) (2018) S33–s39.
[7] V.K. Somers, et al., Sleep apnea and cardiovascular disease: an American heart association/American College of cardiology foundation scientific statement from the American heart association council for high blood pressure research professional education committee, council on clinical cardiology, stroke council, and council on cardiovascular nursing, J. Am. Coll. Cardiol. 52 (8) (2008) 686–717.
[8] T.D. Bradley, J.S. Floras, Obstructive sleep apnoea and its cardiovascular consequences, Lancet 373 (9657) (2009) 82–93.
[9] D.J. Gottlieb, et al., Prospective study of obstructive sleep apnea and incident coronary heart disease and heart failure: the sleep heart health study, Circulation 122 (4) (2010) 352–360.
[10] Y. Yeghiazarians, et al., Obstructive sleep apnea and cardiovascular disease: a scientific statement from the American heart association, Circulation 144 (3) (2021) e56–e67.
[11] L.E. Costa, et al., Potential underdiagnosis of obstructive sleep apnoea in the cardiology outpatient setting, Heart 101 (16) (2015) 1288–1292.
[12] H. Storgaard, et al., At least one in three people with Type 2 diabetes mellitus referred to a diabetes centre has symptomatic obstructive sleep apnoea, Diabet. Med. 31 (11) (2014) 1460–1467.
[13] D.D.M. Nicholl, et al., Declining kidney function increases the prevalence of sleep apnea and nocturnal hypoxia, Chest 141 (6) (2012) 1422–1430.
[14] A.A. Tahrani, et al., Obstructive sleep apnea and diabetic nephropathy: a cohort study, Diabetes Care 36 (11) (2013) 3718–3725.
[15] E. Zamarrón, et al., Obstructive sleep apnea is associated with impaired renal function in patients with diabetic kidney disease, Sci. Rep. 11 (1) (2021) 5675.
[16] J.H. Cho, H.J. Kim, Validation of ApneaLink™ Plus for the diagnosis of sleep apnea, Sleep Breath. 21 (3) (2017) 799–807.
[17] J.M. Fredheim, J. Røislien, J. Hjelmesæth, Validation of a portable monitor for the diagnosis of obstructive sleep apnea in morbidly obese patients, J. Clin. Sleep Med. 10 (7) (2014) 751–757, 757a.
[18] V.K. Kapur, et al., Clinical practice guideline for diagnostic testing for adult obstructive sleep apnea: an American academy of sleep medicine clinical practice guideline, J. Clin. Sleep Med. 13 (3) (2017) 479–504.
[19] Sleep-related breathing disorders in adults: recommendations for syndrome definition and measurement techniques in clinical research. The Report of an American Academy of Sleep Medicine Task Force, Sleep 22 (5) (1999) 667–689.
[20] S. Abbara, et al., SCCT guidelines for the performance and acquisition of coronary computed tomographic angiography: a report of the society of cardiovascular computed tomography guidelines committee: endorsed by the north American society for cardiovascular imaging (NASCI), J Cardiovasc Comput Tomogr 10 (6) (2016) 435–449.
[21] M. Marwan, et al., Very low-dose coronary artery calcium scanning with high-pitch spiral acquisition mode: comparison between 120-kV and 100-kV tube voltage protocols, J Cardiovasc Comput Tomogr 7 (1) (2013) 32–38.
[22] A.S. Agatston, et al., Quantification of coronary artery calcium using ultrafast computed tomography, J. Am. Coll. Cardiol. 15 (4) (1990) 827–832.
[23] R. Detrano, et al., Coronary calcium as a predictor of coronary events in four racial or ethnic groups, N. Engl. J. Med. 358 (13) (2008) 1336–1345.
[24] P. Greenland, et al., Coronary artery calcium score combined with Framingham score for risk prediction in asymptomatic individuals, JAMA 291 (2) (2004) 210–215.
[25] R.L. McClelland, et al., Distribution of coronary artery calcium by race, gender, and age: results from the Multi-Ethnic Study of Atherosclerosis (MESA), Circulation 113 (1) (2006) 30–37.
[26] L.M. Van Bortel, et al., Expert consensus document on the measurement of aortic stiffness in daily practice using carotid-femoral pulse wave velocity, J. Hypertens. 30 (3) (2012) 445–448.
[27] J.N. Luchman, Determining relative importance in Stata using dominance analysis: domin and domme, STATA J. 21 (2) (2021) 510–538.
[28] S. Javaheri, et al., Sleep apnea: types, mechanisms, and clinical cardiovascular consequences, J. Am. Coll. Cardiol. 69 (7) (2017) 841–858.
[29] N.A. Shah, et al., Obstructive sleep apnea as a risk factor for coronary events or cardiovascular death, Sleep Breath. 14 (2) (2010) 131–136.
[30] C.H. Lee, et al., Severe obstructive sleep apnea and outcomes following myocardial infarction, J. Clin. Sleep Med. 7 (6) (2011) 616–621.
[31] G. Weinreich, et al., Association of obstructive sleep apnoea with subclinical coronary atherosclerosis, Atherosclerosis 231 (2) (2013) 191–197.
[32] A.K. Medeiros, et al., Obstructive sleep apnea is independently associated with subclinical coronary atherosclerosis among middle-aged women, Sleep Breath. 21 (1) (2017) 77–83.
[33] D. Sorajja, et al., Independent association between obstructive sleep apnea and subclinical coronary artery disease, Chest 133 (4) (2008) 927–933.
[34] F.S. Luyster, et al., Relation of obstructive sleep apnea to coronary artery calcium in non-obese versus obese men and women aged 45-75 years, Am. J. Cardiol. 114 (11) (2014) 1690–1694.
[35] K.A. Matthews, et al., Associations of Framingham risk score profile and coronary artery calcification with sleep characteristics in middle-aged men and women: pittsburgh SleepSCORE study, Sleep 34 (6) (2011) 711–716.
[36] M.Y. Seo, et al., Association of obstructive sleep apnea with subclinical cardiovascular disease predicted by coronary artery calcium score in asymptomatic subjects, Am. J. Cardiol. 120 (4) (2017) 577–581.
[37] C.J. Porter, et al., Detection of coronary and peripheral artery calcification in patients with chronic kidney disease stages 3 and 4, with and without diabetes, Nephrol. Dial. Transplant. 22 (11) (2007) 3208–3213.
[38] R. Mehrotra, et al., Determinants of coronary artery calcification in diabetics with and without nephropathy, Kidney Int. 66 (5) (2004) 2022–2031.
[39] H.S. Hecht, et al., Coronary artery calcium scanning: clinical paradigms for cardiac risk assessment and treatment, Am. Heart J. 151 (6) (2006) 1139–1146.
[40] H. Nagahama, et al., Pulse wave velocity as an indicator of atherosclerosis in obstructive sleep apnea syndrome patients, Intern Med 43 (3) (2004) 184–188.
[41] C. Tsioufis, et al., The incremental effect of obstructive sleep apnoea syndrome on arterial stiffness in newly diagnosed essential hypertensive subjects, J. Hypertens. 25 (1) (2007) 141–146.
[42] S. Chung, et al., The association of nocturnal hypoxemia with arterial stiffness and endothelial dysfunction in male patients with obstructive sleep apnea syndrome, Respiration 79 (5) (2010) 363–369.

[43] R. Wolk, T. Kara, V.K. Somers, Sleep-disordered breathing and cardiovascular disease, Circulation 108 (1) (2003) 9–12.

[44] R. Tachikawa, et al., Obstructive sleep apnea and abdominal aortic calcification: is there an association independent of comorbid risk factors? Atherosclerosis 241 (1) (2015) 6–11.

[45] S. Kim, et al., Relationship of obstructive sleep apnoea severity and subclinical systemic atherosclerosis, Eur. Respir. J. 55 (2) (2020).

[46] B. Williams, et al., 2018 ESC/ESH guidelines for the management of arterial hypertension: the task Force for the management of arterial hypertension of the European society of cardiology and the European society of hypertension: the task Force for the management of arterial hypertension of the European society of cardiology and the European society of hypertension, J. Hypertens. 36 (10) (2018) 1953–2041.

[47] C. Krogager, et al., Effect of 12 weeks continuous positive airway pressure on day and night arterial stiffness and blood pressure in patients with type 2 diabetes and obstructive sleep apnea: a randomized controlled trial, J. Sleep Res. 29 (4) (2020), e12978.

[48] S.A. Jamal, et al., Effect of calcium-based versus non-calcium-based phosphate binders on mortality in patients with chronic kidney disease: an updated systematic review and meta-analysis, Lancet 382 (9900) (2013) 1268–1277.

[49] B.S. Wojeck, et al., Ertugliflozin and Incident Obstructive Sleep Apnea: an Analysis from the VERTIS CV Trial, Sleep Breath, 2022.

[50] S.S. Ng, et al., Validation of a portable recording device (ApneaLink) for identifying patients with suspected obstructive sleep apnoea syndrome, Intern. Med. J. 39 (11) (2009) 757–762.